

Nutrition Assistant

Setup & Configuration Guide (MERN Stack)

1. Project Overview

The Nutrition Assistant System is a full-stack MERN (MongoDB, Express.js, React.js, Node.js) web application that helps users maintain a healthy lifestyle through personalized diet recommendations, BMI calculation, nutrition tracking, meal planning, and user authentication.

The project contains two modules: Client and Server.

Core Features

User Registration & Login (JWT Authentication)

Personalized Diet Plans

BMI Calculator

Daily Calorie Tracking

Nutrition Recommendation Engine

Meal Planner

Admin Dashboard

Responsive UI

MongoDB Integration

2. Project Folder Structure

Nutrition-Assistant/

Client/

src/

public/

package.json

Server/

config/

controllers/

middleware/

models/

routes/

utils/

server.js

.env

3. Setup Instructions

Create Nutrition-Assistant folder, then Client and Server folders. Open the root folder in VS Code.

Client:

```
npm create vite@latest . -- --template react
npm install
npm install axios react-router-dom react-icons react-toastify chart.js react-chartjs-2
npm run dev
```

Server:

```
npm init -y
npm install express mongoose dotenv cors bcryptjs jsonwebtoken cookie-parser multer
npm install --save-dev nodemon
```

4. Environment Configuration

```
PORT=5000
MONGO_URI=mongodb://localhost:27017/nutritionassistant
JWT_SECRET=nutrition_assistant_secret_key
JWT_EXPIRE=30d
NODE_ENV=development
```

5. Backend Structure

- config/ - Database configuration
- controllers/ - Business logic
- middleware/ - JWT Authentication
- models/ - MongoDB Schemas
- routes/ - API Endpoints
- utils/ - Helper functions
- server.js - Application entry point

6. API Modules

Authentication API

User API

Diet Plan API

Meal API

Nutrition API

BMI API

Water Intake API

Admin API

7. Test Credentials

User: user@nutrition.com / user123

Admin: admin@nutrition.com / admin123

8. Quick Commands

Client: npm run dev

Server: npm run dev

Database: mongod